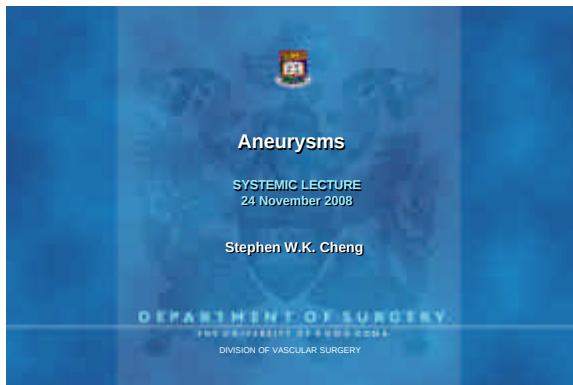
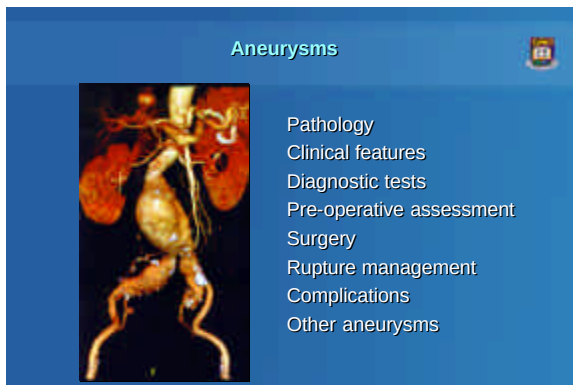
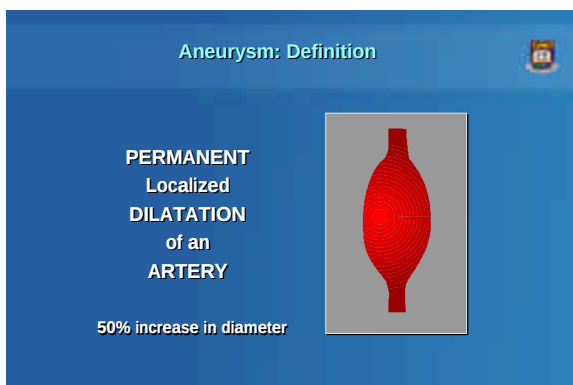
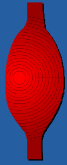








Aneurysm: Classification by Form



FUSIFORM

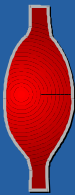


SACCULAR



DISSECTING

Aneurysm: Classification by Structure



TRUE



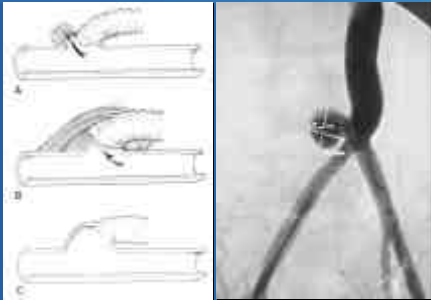
FALSE
(Pseudoaneurysm)

laminated
thrombus
compressed
fibrous tissue

Aneurysm: Etiology



Aneurysm: Anastomotic

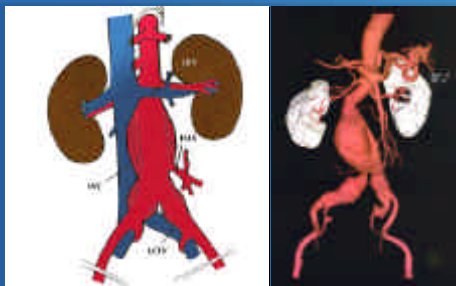


Complications of Aneurysms



- Rupture
- Thrombosis
- Embolism
- Infection
- Pressure effects

Abdominal Aortic Aneurysm (AAA)



Abdominal Aortic Aneurysm (AAA)



Pathology

Loss of elastin and smooth muscle cells
Disruption of extracellular matrix
Deposition of adventitial collagen
Thickening
Inflammatory infiltrate

Abdominal Aortic Aneurysm: Etiology



Multi-factorial

Mechanical: degeneration, BP
Enhancement of proteolytic activity
(Elevated Matrix metalloproteinases (MMP))
Genetic: Marfan's, Ehlers-Danlos IV
Autoimmune
Infection

Abdominal Aortic Aneurysm (AAA)



M>F
97% infrarenal
95% associated atherosclerosis
20% associated aneurysms

AAA: Natural History and Risk of Rupture



Expansion: LaPlace's law: $\sim 5\text{mm} / \text{year}$

Risk of rupture at five years

< 5cm: 20%

> 5cm: 50% (10% per year)

AAA: Clinical Presentation



Asymptomatic

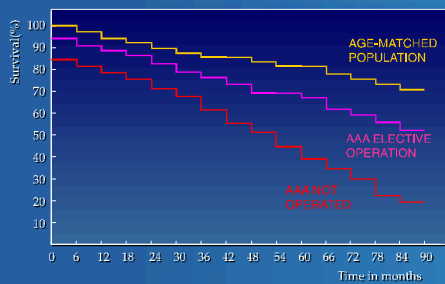
Incidental, PULSATILE abdominal MASS

Symptoms

PAIN: impending rupture



AAA: Survival

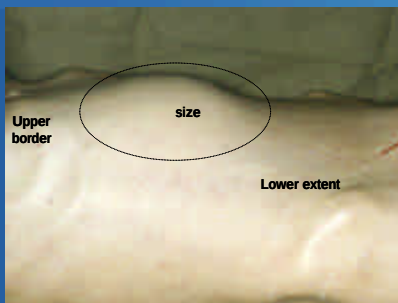


AAA: Physical Examination

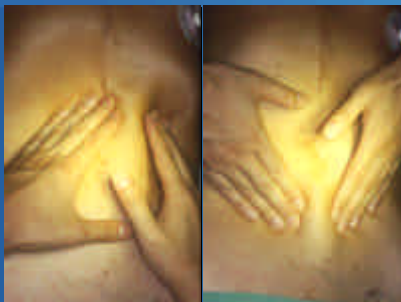


Confirm AAA:	Mass above umbilicus Expansile pulsation
Extent of AAA:	Size Upper border Lower border
Cardiovascular	Pulses, heart, BP

AAA: Physical Examination



AAA: Physical Examination



Plain X-ray calcifications



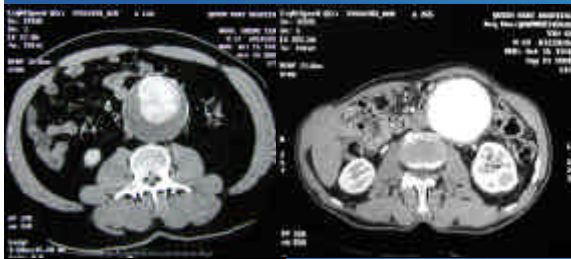
AAA: Calcifications on X-Ray



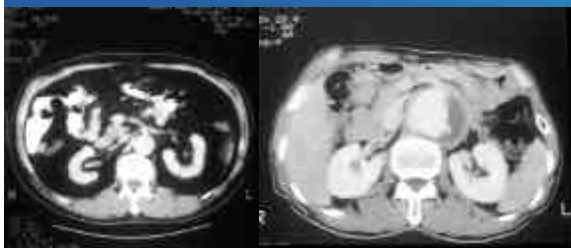
AAA: Ultrasound



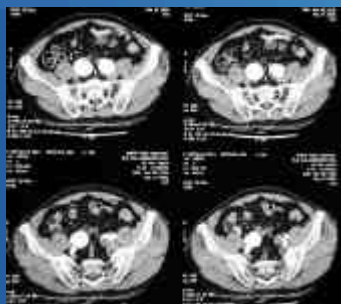
CT Scan



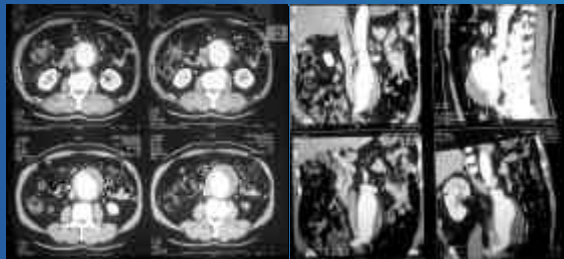
CT Scan: Suprarenal Aneurysm



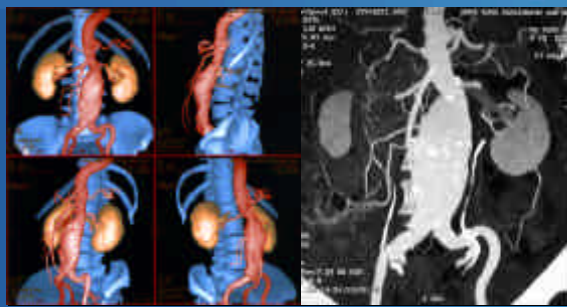
CT Scan: Iliac Aneurysms



AAA: CT Scan



CT Scan



AAA: Operative Considerations



Symptoms:	Any symptoms = urgent
Size:	5cm (approx)
Medical risk:	Associated diseases
Life expectancy:	
Age	Not a contraindication

AAA: Operative Mortality



Risk of OPERATION

Intact aneurysm: 3-5%



Risk of RUPTURE

Ruptured aneurysm: >50%
Unoperated rupture: 100%

All AAA >5cm should be operated on unless patient
Medically unfit (Operative mortality > risk of rupture)
Or limited life expectancy

AAA: Pre-Operative Preparation



General: Blood tests, ECG, CXR

Cardiac: Cardiac assessment / intervention

Preparations: Monitors, blood

Major operative mortality = myocardial infarction

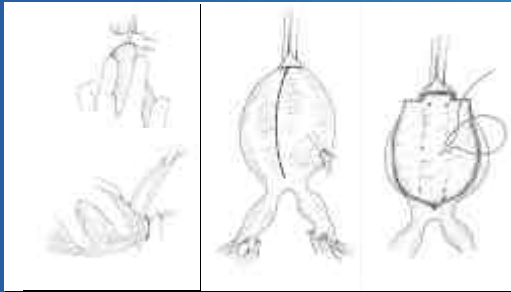
AAA: Surgical Treatment



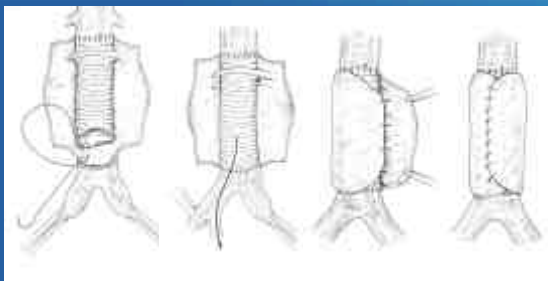
Open Repair: Aneurysmectomy + inlay graft

Endovascular Repair: Aortic stent graft

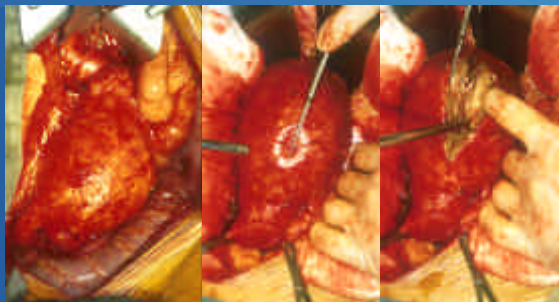
AAA: Open Repair



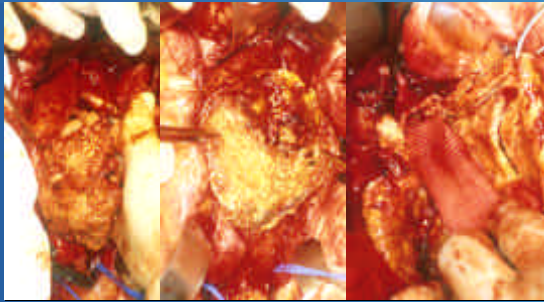
AAA: Open Repair



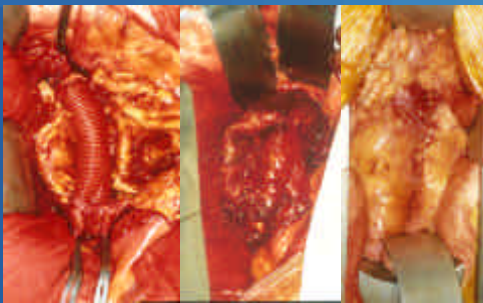
AAA: Open Repair



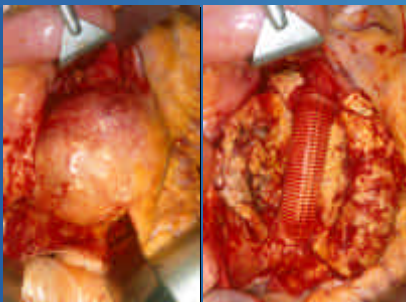
AAA: Open Repair



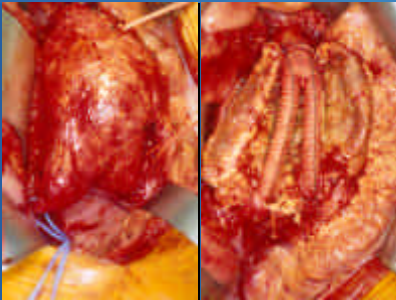
AAA: Open Repair



AAA: Tube Graft



Aorto-iliac Bifurcated Graft



AAA: Operative Complications: Early

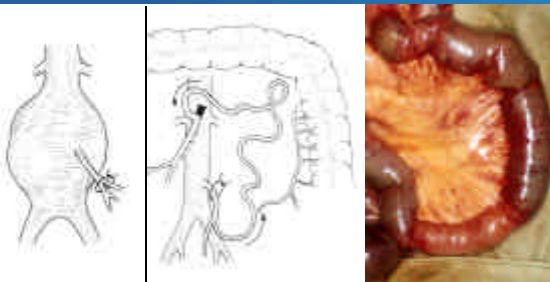


General Complications

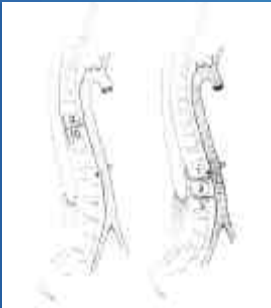
Cardiac (Clamp / declamp)

Respiratory

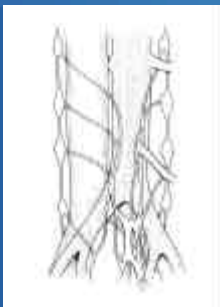
AAA: Operative Complications



AAA: Operative Complications



AAA: Operative Complications



AAA: Operative Complications



Complication: Trash Foot



AAA: Operative Complications: Early



Specific Complications

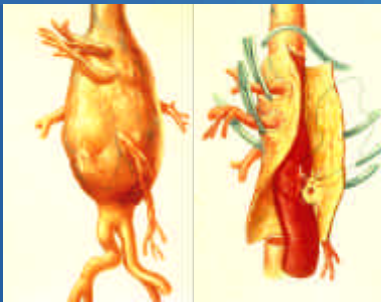
Hemorrhage
Bowel ischemia
Impotence
Renal failure
Distal embolism
Paraplegia

AAA: Operative Complications: Late

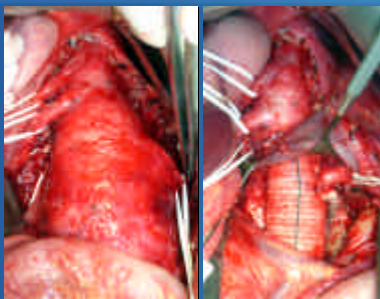


Graft infection
Anastomotic aneurysm
Graft-duodenal fistula

Suprarenal Aneurysms



Suprarenal Aneurysm



Marfan's Thoracoabdominal Dissecting Aneurysm



Suprarenal / Thoracoabdominal Aneurysms

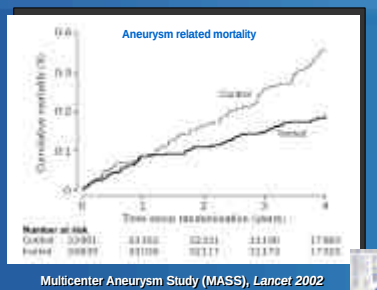


High aortic clamp
Critical ischemic time
Vital branches

Proximal hypertension
Visceral / renal
Spinal ischemia

Bypass
Reimplant visceral arteries

AAA Screening in Men over 65



AAA Screening Reduces Mortality in Men



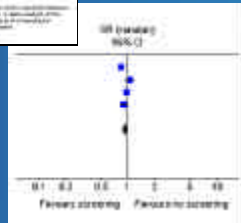
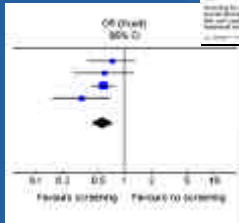
AAA Screening Reduces Mortality in Men



AAA related deaths

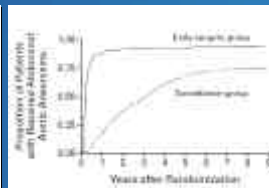
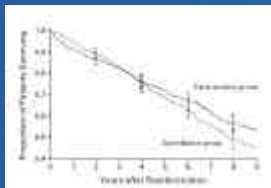


All deaths



Ulinde 2008

Small AAA: Repair or Surveillance?



N=1,090, UK Small Aneurysm Trial (4.0-5.5cm), NEJM 2002



Ruptured AAA



Ruptured AAA



Retroperitoneal

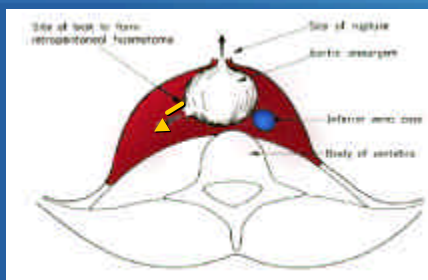
Intraperitoneal (Free)

Into duodenum (GI bleeding)

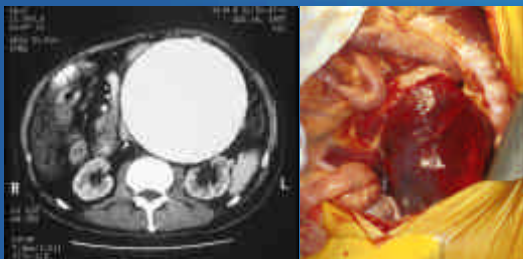
Into IVC (Heart failure)



Ruptured AAA



Aortic Aneurysms



Ruptured AAA: Clinical Features



The TRIAD of Rupture

Pain: Abdomen / back
Mass: Pulsatile (may be masked)
Shock: Transient / profound

Ruptured AAA



Ruptured AAA



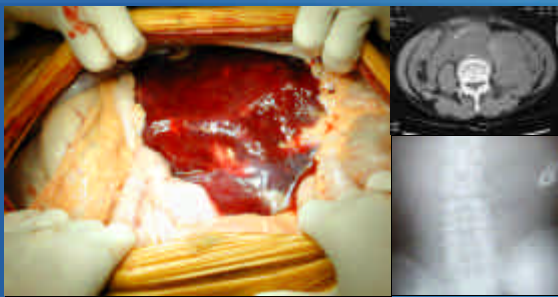
Only 1 in 3 reaches hospital
Surgical emergency
Immediate diagnosis – operation
Operative mortality >50%
Overall mortality >80%

Ruptured AAA: Management

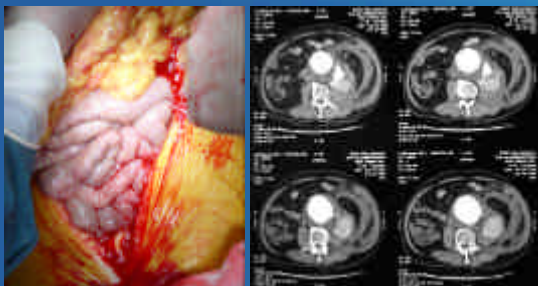


Treat hemorrhagic shock
Large bore IV
Cross match blood / FFP
Immediate operation
Do not waste time in investigations

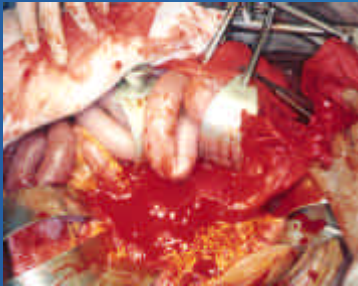
Retroperitoneal Rupture



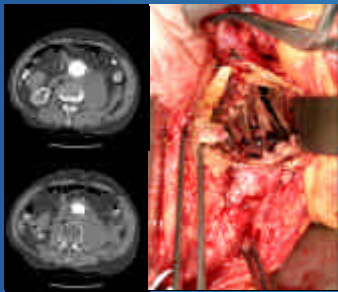
Free AAA rupture



AAA Rupture



Contained Rupture Small AAA

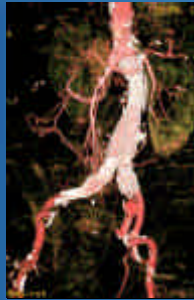


Ruptured AAA: Specific Complications

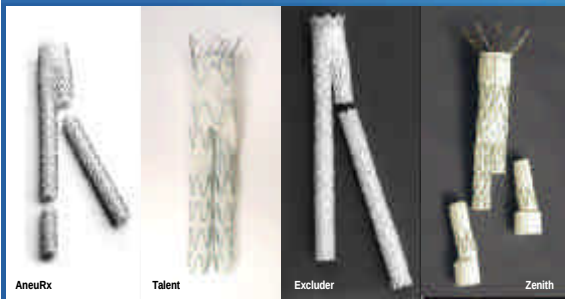


Cardiac	
Respiratory	
Renal failure	(Shock)
Bleeding tendency	(Massive transfusion)
Paralytic ileus	(Retroperitoneal hematoma)
Jaundice	(Bleeding + transfusion)

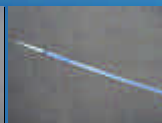
Endovascular Aortic Stent Graft



Aortic Stent Graft Systems



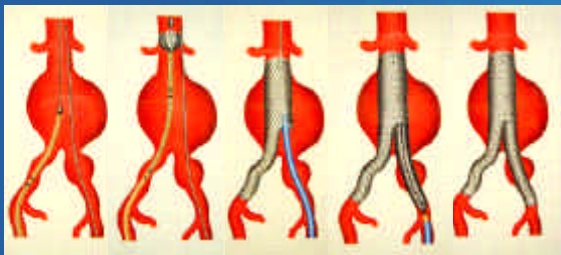
Talent Stent Graft



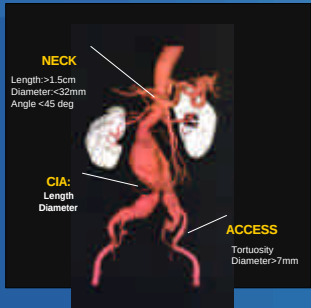
Microport AEGIS



Endovascular Repair for AAA



EVAR: Critical Issues



NECK
Length > 1.5cm
Diameter < 32mm
Angle < 45 deg

CIA:
Length
Diameter

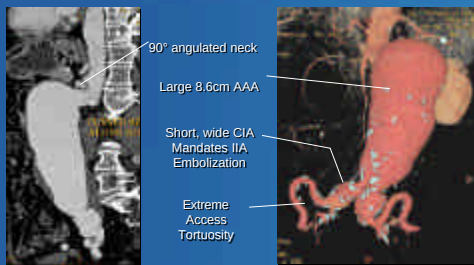
ACCESS
Tortuosity
Diameter > 7mm

Neck: length, angle, diameter
Iliac: Length, diameter
Access
Calcifications
Preserve internal iliac

EVAR: Patient Selection



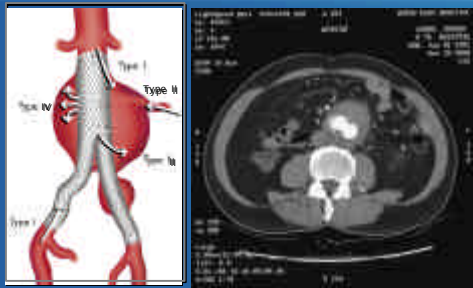
EVAR: Patient Selection

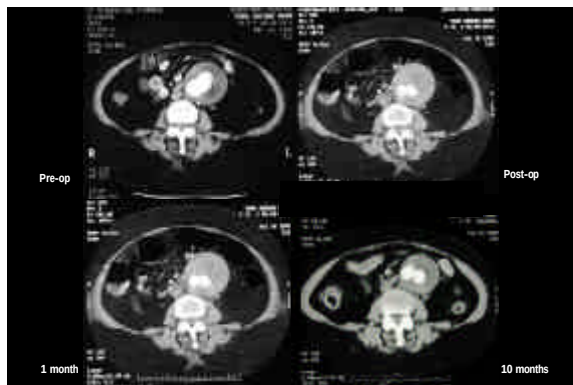


Endovascular Stent Graft for AAA

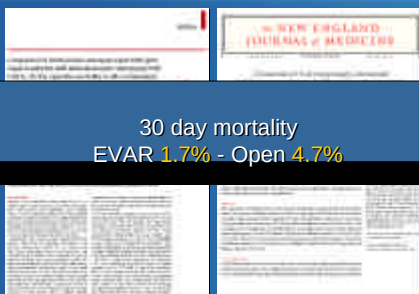


Endoleaks

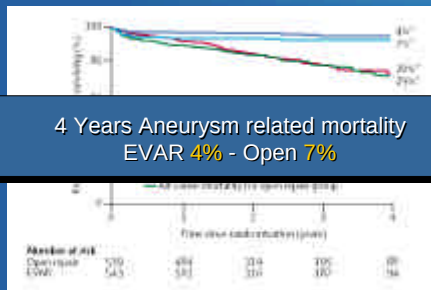




Open vs Endovascular Repair of AAA



EVAR 1: Mortality



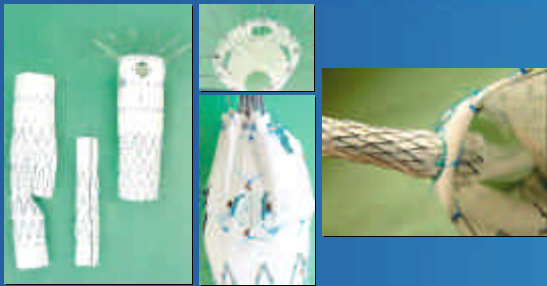
Endovascular Repair: Leaking Aneurysm



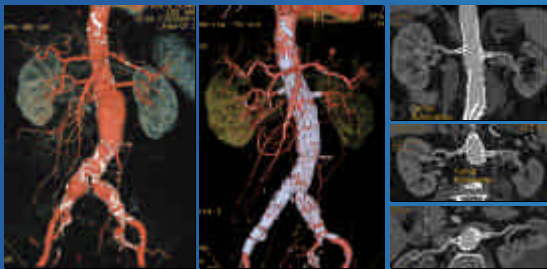
Fenestrated Aortic Stent Graft



Fenestrated Aortic Stent Graft



Fenestrated Aortic Stent Graft



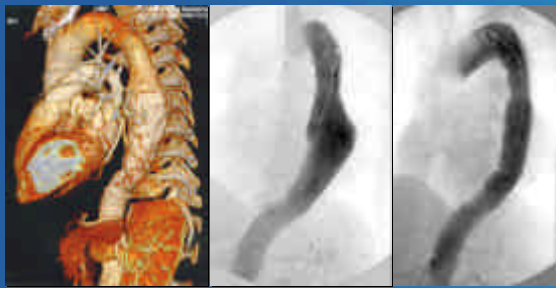
Bifurcated-Bifurcated Graft



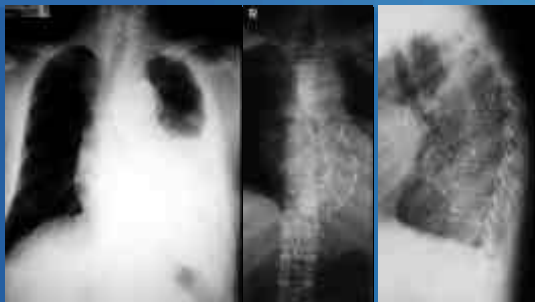
Bifurcated-Bifurcated Graft



Endovascular Repair Thoracic Aneurysms



Thoracic Aortic Aneurysm Rupture



Traumatic Thoracic Aortic Rupture: Endograft



Branched Stent Grafts



Fenestrated Graft



Endovascular Surgery Theatre